

Vessel Risk Classification Report

Automated Workflow Output

MMSI: 123456789
Date: 2025-11-28
Region: Arctic – Beaufort / Chukchi Sector
Classification: Operational Intelligence Product – Unclassified

1. Vessel Overview (from AIS Agent)

Vessel Name: MV Northern Star
Type: Cargo / General
Flag: Liberia
Length / Beam: 145m / 23m

Current Position: 72.44° N, 151.02° W
Speed Over Ground: 8.9 kts
Course Over Ground: 256°
Heading: 248°
Last AIS Update: 4 minutes ago

Track Summary (Past 12 Hours):

- Straight transit west along the Beaufort shelf edge
- Slight deceleration from 11.2 kts → 8.9 kts
- No sharp turns or stops
- No AIS silence / spoofing detected
- No deviations from expected maritime corridor

Behavior Assessment:
Normal. No anomalous patterns detected.

2. AIS Anomaly Analysis (AIS Agent + Planner)

Analyzed Indicators:

- Speed anomalies: **None**
- Heading instability: **Low (normal for region)**
- AIS transmission gaps: **None**
- Unusual course reversals: **None**

- Proximity to restricted zones: **Negative**
- Proximity to other vessels: **>25 NM separation**

AIS Anomaly Score: 1 / 5 (Low)

Interpretation: Vessel behavior consistent with expected transit norms.

3. Environmental Risk Assessment (Prithvi-EO + Prithvi-WxC via METOC Agent)

A. Sea Ice Conditions (Prithvi-EO)

- Ice concentration: **40–55%**
- Ice edge within **12 km** southeast of vessel
- 24-hour drift prediction: **4–7 km westward**
- No major leads or pressure ridges detected

Ice Risk Level: Moderate (3 / 5)

Interpretation:

The vessel is within a dynamic ice zone; prudent watch recommended.

B. Meteorological Conditions (Prithvi-WxC)

- Winds: 18–24 kts from the NE
- Air temp: –19°C
- Seas: 2–3 m
- Visibility: 4–6 km
- Freezing spray index: **Elevated**

Weather Risk Level: Elevated (4 / 5)

Interpretation:

Freezing spray risk is increasing; moderate caution advised.

4. Nearest Ports & Rescue Accessibility (Ports Agent)

Nearest Ports

Port	Distance	Bearing	Status
Utqiagvik	182 km	SSE	Operational
Prudhoe Bay	231 km	SE	Operational
Tuktoyaktuk	312 km	ESE	Operational

Nearest Icebreaker

- *USCGC Healy* (notional)
- Distance: **86 km SW**
- Estimated time to intercept: **5.9 hours**

Port/Support Risk Level: Low (2 / 5)

Interpretation:

Multiple extraction or support options available within operational windows.

5. Operational Risk Thresholds (Knowledge Agent)

Evaluated Against Standard Arctic SOP Thresholds:

Threshold	Status
Ice > 70% concentration	Not exceeded
Freezing spray > High	Exceeded (Moderate/High)
AIS silence > 30 minutes	Not exceeded
Speed instability	Not exceeded
Proximity to ice edge < 10 km	Close (12 km)
Weather > Level 3 severity	Level 3–4

Threshold Assessment:
1 threshold exceeded
2 thresholds near-exceedance
Remaining thresholds nominal

6. Final Vessel Risk Score (Planner Agent Synthesis)

Component Scores:

- AIS Behavioral Risk: **1 / 5**
- Ice Risk: **3 / 5**
- Weather Risk: **4 / 5**
- Logistics / Port Access Risk: **2 / 5**
- SOP Threshold Risk: **3 / 5**

Final Classification: MODERATE RISK

(Yellow / Elevated Watch Level)

Rationale (Explainability Layer):

- Vessel behavior is normal, but
- Weather and freezing spray conditions pose significant operational risk
- Ice edge proximity is within a drift window
- One SOP threshold exceeded. Two approaching yellow

Recommended Actions:

1. Maintain elevated watch on vessel over next 6–10 hours.
2. Refresh ice and weather model runs every 3–4 hours.
3. Notify regional watch if freezing spray index increases.
4. No rerouting recommendation currently.

7. Automated SITREP (Report Agent Output)

SITREP – Arctic Vessel Monitoring

Subject Vessel: MMSI 123456789

Classification: Moderate Risk

Time: 2025-11-28 15:00Z

Summary:

MV Northern Star shows normal transit behavior, but environmental conditions elevate operational risk. Ice edge proximity and increasing freezing spray conditions warrant continued monitoring.